Course-End Project

AI-Powered LinkedIn Content Automation with n8n and AutoGen Microservice

Overview

This project challenges you to design and implement an AI-powered workflow automation system that streamlines LinkedIn content creation and posting for a fintech organization. The solution combines an AutoGen-inspired microservice with the n8n workflow orchestrator. The workflow demonstrates intelligent orchestration across multiple AI-driven components, including side ideation, drafting, and hashtag generation, before applying guardrails for approval and publishing. The project highlights how product leaders can deploy automation to achieve efficiency, brand consistency, and scale in content strategy.

Instructions

- Review the lessons and supporting materials on n8n workflows and AutoGen-style multi-agent design
- Set up the required environment on the Ubuntu VM, including Node.js, n8n, and the FastAPI microservice
- Follow step-by-step development to:
    - Build the AutoGen-style microservice (/linkedin endpoint)
    - Configure and connect the n8n workflow with Brand Config, AutoGen Microservice, Compose Final, and Approval Gate nodes
    - Implement Slack for dry-run testing and prepare LinkedIn integration for live posting
- Test and debug each component individually (microservice with curl, each n8n node with Execute Node, and then the full workflow)
- Document the architecture, configuration steps, test runs, and error resolutions
- Submit:
    - Microservice code (main.py, .env.example)
    - Exported n8n workflow JSON
    - Screenshots of successful runs
    - A short reflection on design decisions, challenges, and trade-offs

Situation

FinEdge Mumbai, a fintech company with over five hundred employees, is facing significant challenges in its LinkedIn strategy:

- The content team spends 15+ hours per week manually drafting posts.
- Inconsistent posting has reduced engagement by 45%.
- Manual content creation is too slow, causing posts to miss trending topics.
- Executives lack a unified voice, weakening brand authority.

Despite having rich domain expertise, the company struggles to maintain an authoritative LinkedIn presence. To overcome this, the product team decides to implement an AutoGen-style multi-agent workflow that generates consistent, on-brand LinkedIn posts with minimal human effort, while retaining oversight through approval controls.

Tasks

- Build a multi-agent microservice that returns multiple post ideas, a draft post with confidence scoring, and relevant hashtags
- Deploy and test the microservice on your VM, ensuring it accepts brand and context information via an API and returns structured JSON output
- Configure an n8n workflow with nodes for scheduling, brand configuration, microservice invocation, composing the final post, approval gate, and routing to Slack or LinkedIn
- Include mechanisms for logging and error handling to ensure transparency and traceability

Actions

To complete this project, you will have to:

- Design and implement the AutoGen-style microservice using FastAPI, exposing an endpoint that accepts brand configuration and context, then returns JSON containing ideas, draft, confidence score, and hashtags
- Run and test the microservice locally using curl or HTTP client tools to ensure it produces the desired output format
- Create a new workflow in the n8n dashboard, with a schedule trigger set to a suitable cadence for LinkedIn posting
- Add nodes for brand configuration and pass this data to the microservice using an HTTP Request node with JSON parameters
- Parse the microservice response to assemble the final text and hashtags using a Set node (no JavaScript required)
- Implement an approval gate that compares the confidence score against a minimum threshold and checks a dry-run flag to route posts to Slack for review or LinkedIn for direct publishing
- Configure Slack integration via incoming webhook or OAuth (optional) to send draft posts for human review
- Add logging by appending run details (for example, timestamp, draft text, confidence) to a database or spreadsheet to monitor performance and tune thresholds
- Test the complete flow end-to-end and adjust parameters (for example, confidence threshold) for optimal balance between automation and oversight

Result

By the end of this project, you will deliver a working n8n workflow that automates LinkedIn content generation, a functioning AutoGen-style microservice, a thorough README detailing setup and run instructions, and documentation summarizing your design decisions and testing outcomes.